

MICROWAVE OVEN SIMULATION

1. Introduction

The Microwave Oven Simulation project is an embedded systems-based application developed to simulate the working of a microwave oven controller. The project focuses on implementing the core control logic of a microwave oven using a microcontroller-based approach. It demonstrates how different cooking modes, user inputs, timing control, and display units work together in a real-time embedded system.

This project was developed as part of an Embedded Systems Internship at Emertxe and is intended for academic learning and demonstration purposes.

2. Project Objective

The main objectives of this project are:

- To simulate the functionality of a microwave oven controller
 - To implement multiple operating modes of a microwave oven
 - To understand timer-based and interrupt-driven control
 - To design a modular embedded software architecture
 - To demonstrate user interaction using keypad and LCD
-

3. Project Overview

The project simulates a microwave oven that allows the user to select different cooking modes and set cooking time. Based on the selected mode, the system controls timing and displays relevant information on an LCD. The complete operation is managed using a state-based control logic with the help of timer interrupts.

The simulation includes four modes:

- Micro Mode
- Grill Mode
- Convection Mode
- Start Mode

4. System Architecture

The system architecture consists of the following main blocks:

- **User Input Unit:** Matrix keypad used for mode selection and time setting
- **Control Unit:** Microcontroller that processes inputs and controls system flow
- **Timing Unit:** Timer and interrupt mechanism for accurate time management
- **Display Unit:** Character LCD to show mode, time, and system status

The microcontroller acts as the central unit that connects all peripherals and executes the control logic.

5. Operating Modes

5.1 Micro Mode

- Used for normal cooking operation
- Simulates high-power microwave heating
- User sets cooking time manually

5.2 Grill Mode

- Used for grilling food items
- Similar to micro mode but intended for surface heating

5.3 Convection Mode

- Used for baking and roasting operations
- Includes a preheating phase
- Maximum temperature limited to 180°C
- After preheating, cooking time can be set by the user

5.4 Start Mode

- Default quick-start mode
- Automatically starts cooking for a fixed short duration

6. Hardware and Software Requirements

6.1 Software Tools

- MPLAB X IDE
- XC8 Compiler
- PICSimLab

6.2 Hardware / Simulation Components

- PIC16F877A Microcontroller
 - 16×4 Character LCD (CLCD)
 - Matrix Keypad
 - Timer2 (Internal Peripheral)
 - Power Supply
-

7. Module Description

7.1 LCD Module (CLCD)

The 16×4 Character LCD is used to display system messages such as selected mode, cooking time, and system status. It provides user feedback and improves system interaction.

7.2 Matrix Keypad

The matrix keypad is used for user input. It allows the user to select operating modes and set cooking time. Key detection is done using row and column scanning.

7.3 Timer and ISR

Timer2 is used to generate periodic interrupts. The Interrupt Service Routine (ISR) handles time counting and controls the cooking duration accurately without blocking the main program.

8. Software Design

The software is developed using a modular approach. Each peripheral has a separate source and header file. The main program initializes all modules and controls the overall flow using a state machine.

Key features of the software design include:

- Modular programming
 - State-based control logic
 - Interrupt-driven timing
 - Reusable driver modules
-

9. Project Implementation

The project implementation involves initializing all peripherals, detecting keypad input, updating the LCD, and managing time using timer interrupts. Based on the selected mode, the system transitions between different states until the cooking operation is completed.

10. Applications

- Learning embedded systems concepts
 - Academic mini-projects
 - Embedded systems laboratory demonstrations
 - Understanding real-time control systems
-

11. Limitations

- This project is a simulation and does not include power electronics
 - No actual heating or temperature control is implemented
 - Intended only for learning and demonstration purposes
-

12. Conclusion

The Microwave Oven Simulation project successfully demonstrates the working of a microwave oven controller using embedded systems concepts. It provides practical exposure to timers, interrupts, peripheral interfacing, and modular programming. The project serves as a strong foundation for understanding real-world embedded control systems.

13. Future Enhancements

- Adding temperature sensors for real-time feedback
 - Implementing buzzer alerts
 - Adding more cooking modes
 - Integrating serial communication or IoT features
-

14. References

- PIC16F877A Datasheet
- Embedded C Programming Concepts
- Emertxe Embedded Systems Training Material